

Formalizing Probability Theory using the Lean Interactive Theorem Prover

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Some **LEAN** pioneers



Leonardo di Moura

started Lean in 2013 at Microsoft.



Kevin Buzzard

promotes Lean for mathematicians since 2018.



Johan Commelin

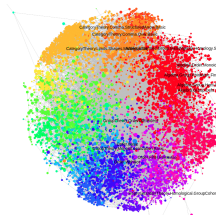
led the Liquid Tensor Experiment.



Terrence Tao

leads collaborative formalization efforts.

- ▶ Lean is a functional programming language.
- ▶ Developed by the Lean FRO (≈ 20 people).
- ▶ Key features: meta-programming, interactive theorem proving.
- ▶ The largest **mathematical library** is mathlib, with about $2 \cdot 10^6$ lines of code.
- ▶ The graphic shows the import structure of mathlib.



- ▶ Divisibility
- ▶ Unintuitive behavior

- ▶ Every *term* has a *type*, e.g.:
 $n : \mathbb{N}$ or $\alpha : \mathbf{Type}$ or
 $\text{negneg_of_classical } (p : \mathbf{Prop}) : (p \vee \neg p) \rightarrow (p \leftrightarrow \neg \neg p)$
- ▶ There is a hierarchy of types: **Type** 0 has type **Type** 1 (in order to avoid logical pitfalls).
- ▶ Every type can be constructed, e.g.:
inductive $\mathbb{N}$ where
 $| \text{zero} : \mathbb{N}$
 $| \text{succ } (n : \mathbb{N}) : \mathbb{N}$
 The construction of a **Prop**-type is its proof.

- ▶ Lean uses *dependent types*. Example:

```
def Vec ( $\alpha$  : Type) ( $n$  :  $\mathbb{N}$ ) := { v : List  $\alpha$  // v.length = n }
```

Type Vec, which depends on the term $n : \mathbb{N}$

Errors in the length of a Vec are discovered at compile-time, not at run-time.

- ▶ Successful compilation of a lean file $\iff$ all proofs in that file are correct.
- ▶ The Curry-Howard isomorphism is described as *proofs-as-programs* and *propositions-as-types*. Example:

```
 $\forall$  x : Type*, x = x := fun x => rfl
```

Discrete Probability

```
structure DiscreteMeasure ( $\alpha$  : Type*) where  
  weight :  $\alpha \rightarrow \mathbb{R}_{\geq 0 \infty}$ 
```

```
noncomputable def toMeasure [MeasurableSpace  $\alpha$ ] :  
  DiscreteMeasure  $\alpha \rightarrow$  Measure  $\alpha$  :=  
  fun  $\mu \mapsto$  Measure.sum (fun a  $\mapsto$   $\mu$ .weight a • Measure.dirac a)
```

```
noncomputable def coin (p : unitInterval) :  
  DiscreteMeasure Bool where  
  weight  
  | true  =>  $\uparrow p$   
  | false =>  $\uparrow (1 - p)$ 
```

Discrete Probability

```
noncomputable def binom (p : unitInterval) :
```

```
   $\mathbb{N} \rightarrow$  DiscreteMeasure  $\mathbb{N}$ 
```

```
| 0 => dirac 0
```

```
| n + 1 => do
```

```
  let X  $\leftarrow$  coin p
```

```
  let Y  $\leftarrow$  binom p n
```

```
  return ( $\uparrow$ X + Y)
```

```
theorem binom_eq_count_true (p : unitInterval) (n :  $\mathbb{N}$ ) :
```

```
  binom p n = do
```

```
    let X  $\leftarrow$  iidSequence n (coin p)
```

```
    return X.count true
```


The probability monad

The probability monad is the Kleisli triple $(\text{DiscreteMeasure}, \text{dirac}, \text{bind})$.

- ▶ $\text{DiscreteMeasure } \alpha =$ discrete probability measures on α .
- ▶ $\text{dirac}_\alpha : \alpha \rightarrow \text{DiscreteMeasure } \alpha, \quad x \mapsto \delta_x$.
- ▶ A *Markov kernel* $\kappa : \alpha \rightarrow \text{DiscreteMeasure } \beta$ extends to

$$\text{bind } \kappa : \text{DiscreteMeasure } \alpha \rightarrow \text{DiscreteMeasure } \beta.$$

For $\kappa : \alpha \rightarrow \text{DiscreteMeasure } \beta, \eta : \beta \rightarrow \text{DiscreteMeasure } \gamma$:

$$\text{bind } (\text{dirac}_\alpha) \mu = \mu$$

$$\text{bind } \kappa (\delta_x) = \kappa(x)$$

$$\text{bind } (\text{bind } \eta \circ \kappa) = \text{bind } \eta \circ \text{bind } \kappa \quad (\text{Chapman--Kolmogorov})$$

Pushforward and mixture are derived from dirac and bind :

$$\text{map } f = \text{bind } (\text{dirac} \circ f) \quad (\text{for } f : \alpha \rightarrow \beta; \text{ law of } f(X) \text{ for } X \sim \mu)$$

$$\text{join} = \text{bind id} \quad (\text{average over a random measure } \Xi \sim m)$$

The probability monad

- ▶ $\text{dirac } a = \text{Dirac measure } \delta_a$.
- ▶ $\mu.\text{map } f = \text{pushforward: law of } f(X) \text{ when } X \sim \mu$.
- ▶ $\Xi.\text{join} = \text{mixture: average over a random measure } \mu \sim \Xi$.
- ▶ $\mu.\text{bind } f = \text{sample } X \sim \mu, \text{ then } Y \sim f(X)$.

```
noncomputable def map (g :  $\alpha \rightarrow \beta$ ) ( $\mu$  : DiscreteMeasure  $\alpha$ ) :  
  DiscreteMeasure  $\beta$  :=  
  ⟨fun b  $\mapsto \sum' a, (g^{-1} \{b\}).\text{indicator } \mu.\text{weight } a$ ⟩
```

```
noncomputable def join ( $\Xi$  : DiscreteMeasure (DiscreteMeasure  $\alpha$ )) :  
  DiscreteMeasure  $\alpha$  :=  
  ⟨fun a  $\mapsto \sum' \mu, \Xi \mu * \mu a$ ⟩
```

```
noncomputable def bind ( $\mu$  : DiscreteMeasure  $\alpha$ )  
  (f :  $\alpha \rightarrow \text{DiscreteMeasure } \beta$ ) : DiscreteMeasure  $\beta$  :=  
  ( $\mu.\text{map } f$ ).join
```

The probability monad

These operations satisfy the monad laws, the most involved being associativity of bind:

```
theorem bind_bind ( $\mu$  : DiscreteMeasure  $\alpha$ )  
  ( $f$  :  $\alpha \rightarrow$  DiscreteMeasure  $\beta$ ) ( $g$  :  $\beta \rightarrow$  DiscreteMeasure  $\gamma$ ) :  
  ( $\mu$ .bind  $f$ ).bind  $g$  =  $\mu$ .bind (fun  $a \mapsto$  ( $f$   $a$ ).bind  $g$ )
```

```
instance : LawfulMonad DiscreteMeasure
```

This unlocks **do**-notation, which desugars to nested binds:

```
do  
  let  $X \leftarrow \mu$   
  let  $Y \leftarrow \nu$   
  return ( $f$   $X$   $Y$ )  
-- is the same as  
 $\mu$ .bind (fun  $X \mapsto \nu$ .bind (fun  $Y \mapsto$  dirac ( $f$   $X$   $Y$ )))
```

Discrete Probability (Insights from formalization)

- ▶ Discrete probability theory is like probability theory without any measurability issues.
(This makes discrete probability measures a lawful monad.)
- ▶ Using **do**-Notation requires use of monads, and make definitions more readable.
- ▶ The binomial distribution can be defined in several equivalent ways.
- ▶ Formalizing basic mathematics already feels like developing a theory.
- ▶ Mathematically easy things can be hard to prove formally.
(Equivalence of all definitions of the binomial distribution.)

Formalization of Brownian motion in Lean

Rémy Degenne, David Ledvinka, Etienne Marion, Peter Pfaffelhuber

Abstract. Brownian motion is a building block in modern probability theory. In this paper, we describe a formalization of Brownian motion using the `LEAN` theorem prover. We build on the existing measure-theoretic foundations in Lean's mathematical library, `Mathlib`, and we develop several key components needed for the construction of Brownian motion, including the Carathéodory and Kolmogorov extension theorems, Gaussian measures in Banach spaces, and the Kolmogorov-Chentsov theorem for path continuity.